Looking Within: Gender Quotas and the Composition of Municipal Education Spending in Spain

APEP Autonomous Research*

March 2, 2026

Abstract

Spain's 2007 Equality Law imposed gender quotas on municipal candidate lists above population thresholds, yet researchers have found no effects on aggregate spending. I decompose this aggregate null using an election-term-level regression discontinuity design that matches each council to its election-year population. The 5,000-inhabitant threshold bundles the quota with a council size increase, and the net first-stage effect on female share is negative in key cohorts—consistent with seat dilution offsetting quota compliance. Despite this, the threshold produces detectable reduced-form effects on spending composition. Before Spain's 2013 fiscal reform (LRSAL), municipalities above the threshold allocated 7.5 percentage points more of their education budget to primary school facilities ($p = 0.008$; Benjamini-Hochberg $q = 0.050$). After the reform restricted municipal discretion, this effect reversed. The full-sample null reflects cancellation of opposing period-specific effects. Institutions, not preferences, explain the European null.

JEL Codes: D72, H72, H75, J16

Keywords: gender quotas, political representation, municipal spending, education composition, regression discontinuity

*Autonomous Policy Evaluation Project. Correspondence: scl@econ.uzh.ch

1. Introduction

On the morning of May 28, 2007, Spain's Electoral Board rejected hundreds of candidate lists. The new Equality Law required at least 40% women on every municipal slate above 5,000 inhabitants—and the Board enforced it mechanically, with no appeals. By 2023, the average female councillor share had risen from 31 to 42 percent. But when researchers looked for fiscal consequences, they found nothing: Bagues and Campa (2021) document that the same Spanish quota thresholds produce no effects on aggregate spending categories. The result joins a growing collection of European nulls—from Italian municipalities (Gagliarducci and Paserman, 2012) to U.S. cities (Ferreira and Gyourko, 2014)—that together constitute the central puzzle in the gender and public finance literature: quotas change who governs, but not what governments spend.

This paper explains the puzzle. The answer is not that representation does not matter—it is that institutions determine whether representation *can* matter. I open the black box of municipal education budgets, decomposing "education spending" into eight three-digit program categories, and find that the aggregate null masks a sharp temporal pattern. Before Spain's 2013 fiscal reform (*Ley de Racionalización y Sostenibilidad de la Administración Local*, LRSAL), municipalities above the quota threshold allocated 7.5 percentage points more of their education budget to primary school facilities (program 321; $p = 0.008$). This finding survives Benjamini-Hochberg correction for multiple testing across all education subcategories ($q = 0.050$). After the LRSAL restricted municipal competences, the effect reversed: municipalities above the threshold shifted *away* from primary school spending. The full-sample null is not the absence of effects—it is the cancellation of opposing period-specific effects.

I refine the measurement of the running variable. While previous work averaged population over years, I match each council to its exact election-year population—the figure the Electoral Board actually uses to determine quota applicability. This election-term design also allows decomposition of the first stage by election cohort, revealing how the threshold's compliance effect evolves across the five elections from 2007 to 2023.

Three findings emerge. First, the threshold effect on council composition varies across elections. At 5,000 inhabitants, the threshold bundles the quota with a council size increase from 11 to 13 seats, so the net discontinuity in female share can be negative (seat dilution exceeds quota gains). The 2011 election shows the largest discontinuity (-0.093, $p < 0.001$); by 2015 and 2019, voluntary compliance has eliminated the treatment contrast. This "shelf life" pattern warns that the estimand in multi-election threshold RDD designs is not stable over time.

Second, the pre-LRSAL compositional effect is the paper's central finding. In the 2007 and 2011 election terms—when institutions permitted discretion and the threshold effect was strongest—municipalities above 5,000 allocated 7.5 percentage points more of their education budget to primary school facilities: renovating kitchens, adding accessibility features, and building nursery wings rather than funding administrative overhead. The effect, equivalent to roughly EUR 10 per capita, is driven by the 2011 cohort where the running variable is exact ($+0.080$, $p = 0.051$). Levels and extensive-margin specifications confirm that this operates through reallocation within existing budget lines, not the creation of new spending programs.

Third, the post-LRSAL reversal is itself informative. After 2013, the same threshold is associated with *less* primary school spending (-0.051, $p = 0.025$). This sign reversal is consistent with the LRSAL having fundamentally altered the decision environment: when fiscal rules imposed discipline, the residual discretion may have been exercised differently, or municipalities above the threshold—with larger, more constrained councils—may have responded more sharply to the reform.

The paper contributes to three literatures. First, it reinterprets the gender and public finance null. Where previous work concluded that quotas do not affect spending (Ferreira and Gyourko, 2014; Bagues and Campa, 2021), I show that quotas affect spending composition *conditional on institutional context*. The binding constraint is not the absence of gendered preferences but the institutional channeling of those preferences. This interpretation generates testable predictions: quota effects should appear in settings with greater local fiscal autonomy (Nordic municipalities, Swiss cantons) and in expenditure categories with greater discretionary content.

Second, the paper demonstrates the value of within-category budget analysis. A growing literature examines quotas' effects on candidate quality (Baltrunaite et al., 2014; Besley et al., 2017), legislative behavior (Washington, 2008; Bagues and Esteve-Volart, 2022), and social outcomes (Bhalotra and Clots-Figueras, 2014; Iyer et al., 2012). This paper shows that the natural "next step" in the measurement hierarchy—decomposing spending categories—can reveal effects invisible at the aggregate level, provided the analyst accounts for temporal heterogeneity driven by institutional change.

Third, the election-term design and by-cohort first-stage decomposition offer a methodological template for multi-election RDD studies. The decay of the threshold effect across elections—from significant in 2011 to null by 2019—demonstrates that pooling across elections can mask temporal variation in the treatment contrast. Future work using quota thresholds should decompose by election cohort and use election-year-specific running variables.

2. Institutional Background

2.1 Gender Quotas in Spanish Municipal Elections

Spain introduced mandatory gender quotas for electoral lists through the *Ley Orgánica 3/2007 para la igualdad efectiva de mujeres y hombres*, commonly known as the Equality Law (Boletín Oficial del Estado, 2007). Article 44 bis of the amended Electoral Law (*Ley Orgánica del Régimen Electoral General*) requires that candidate lists for all elections maintain a minimum representation of 40 percent for each gender, in both the full list and in every group of five candidates.

Crucially for identification, the law applied only to municipalities with more than 5,000 registered inhabitants. Smaller municipalities were exempted because their councils have fewer seats (typically 7–9), making strict list composition rules impractical. The 5,000-inhabitant threshold thus created a sharp discontinuity in the regulatory environment faced by political parties when composing their lists.

In 2011, *Ley Orgánica 2/2011* extended the quota requirement to municipalities with more than 3,000 inhabitants (Boletín Oficial del Estado, 2011). This second threshold provides an additional cutoff for the RDD analysis, applicable from the 2011 municipal elections onward.

Enforcement is mechanical: the Junta Electoral (Electoral Board) reviews candidate lists before the election and rejects any list that does not comply with the gender balance requirement. Parties whose lists are rejected must resubmit compliant lists or forfeit their candidacy. This enforcement mechanism ensures near-perfect compliance with the list composition requirement among municipalities above the threshold. However, as shown in Section 5.2, the empirical first stage on *elected* female representation is weak in the full 2007–2023 sample, because parties below the threshold voluntarily adopted balanced lists—complying with the spirit of the law even where it was not legally binding.

Spanish municipal councils (*ayuntamientos*) are elected every four years through a proportional representation system (D'Hondt method). The council size depends on population: municipalities with 251–1,000 inhabitants elect 7 councillors, those with 1,001–2,000 elect 9, those with 2,001–5,000 elect 11, those with 5,001–10,000 elect 13, and those with 10,001–20,000 elect 17. The council elects the mayor (*alcalde/sa*) from among its members. Importantly for this study, fiscal authority rests with the full council through budget approval, not solely with the mayor—making council composition the relevant measure of political representation.

2.2 Municipal Budget Classification

Spanish municipal budgets follow a standardized classification system established by *Orden EHA/3565/2008* (Ministerio de Economía y Hacienda, 2008), which reformed the budget structure for all local entities beginning with the 2010 fiscal year. This reform introduced a three-level program classification (*clasificación por programas*) that disaggregates spending into policy areas, policy groups, and specific programs.

Education spending falls under Policy Area 3 ("Production of Public Goods of a Preferential Nature"), Policy Group 32 ("Education"). Within this group, the three-digit program codes distinguish between:

- **Program 320:** General administration of education.
- **Program 321:** Primary school facilities—construction, maintenance, and administration of educational centers (primarily infant and primary education).
- **Program 322:** Secondary education facilities and services.
- **Program 323:** Education promotion—operational support for schools, including material provision and educational programming.
- **Program 324:** Non-regulated education—adult education, music schools, and language programs.
- **Program 326:** Complementary services—school meals (*comedores escolares*), school transport, extracurricular activities, and summer programs.
- **Programs 325, 327:** University education and special education.

Some municipalities report a portion of their education spending at the aggregate two-digit level (program group 32) without disaggregating into three-digit codes. I calculate within-education shares over the disaggregated three-digit programs only, excluding the residual aggregate code.

This program-level classification is the key data innovation of this paper. All municipalities, regardless of size, report their executed budgets (*liquidaciones*) to the Ministry of Finance using this classification. The data are compiled in the CONPREL database and made publicly available. Unlike France, where functional budget classification was mandatory only for larger communes, Spain's system applies universally—making within-category analysis feasible on both sides of the population thresholds.

2.3 The LRSAL Fiscal Reform

In December 2013, Spain enacted the *Ley de Racionalización y Sostenibilidad de la Administración Local* (LRSAL), a sweeping reform of local government prompted by the fiscal consequences of the Great Recession and European austerity requirements (Boletín Oficial del Estado, 2013). The LRSAL clarified and restricted municipal competences (*competencias propias*), particularly for smaller municipalities. Education was one area affected: the law sought to rationalize which level of government (municipal, provincial, or autonomous community) was responsible for which services.

The LRSAL is relevant for this study in two ways. First, it introduces potential heterogeneity: the scope for municipal reallocation within education may differ before and after 2014. Second, it provides a natural check: if the quota effects operate through council deliberation over discretionary allocations, they should be stronger in the pre-LRSAL period when municipalities had greater autonomy over education spending. I examine this directly in the heterogeneity analysis.

2.4 Municipal Education Competences

Understanding the scope for political influence over education spending requires clarifying what Spanish municipalities actually control. Under the pre-LRSAL framework (*Ley 7/1985 Reguladora de las Bases del Régimen Local*), municipalities had both mandatory and voluntary competences in education. Mandatory competences included: providing land for public schools, maintaining school buildings, and cooperating with the autonomous community's education authority in school construction and renovation. Voluntary competences—which municipalities could choose to exercise—included operating school meal programs (*comedores escolares*), providing school transport, organizing extracurricular activities, funding adult education and music schools, and supporting nursery (*escuelas infantiles*) services.

This distinction between mandatory and voluntary competences maps directly onto the budget program classification. Programs 321 (school facility administration) and 322 (secondary education) reflect largely mandatory obligations—capital expenditures on school buildings that municipalities must provide regardless of council preferences. Programs 324 (non-regulated education) and 326 (complementary services) reflect voluntary competences where councils have genuine discretion: a municipality can choose to operate a school meal program or not, to subsidize music education or not, to provide school bus routes or not.

The compositional hypothesis thus has a clear prediction: if female councillors have different preferences over public goods, those preferences should manifest primarily in the discretionary programs (324, 326, and to some extent 323), not in the mandatory capital

expenditures (321, 322). This is a stronger prediction than the generic "reallocation" story, because it identifies the institutional mechanism through which preferences translate into budgets.

The LRSAL disrupted this framework by narrowing voluntary competences for smaller municipalities. Under the new law, municipalities below 20,000 inhabitants could not exercise competences "impropia" (non-statutory functions) unless they demonstrated financial sustainability. Many smaller municipalities responded by transferring voluntary education services to the provincial council (*diputación provincial*) or the autonomous community. This institutional change provides the source of temporal heterogeneity explored in Section 6.

2.5 Spanish Municipal Elections and the Gender Gap

Spain's municipal election calendar provides clean periodicity for the analysis. Municipal elections were held in May 2003, 2007, 2011, 2015, 2019, and 2023. Each election produces a council that governs for the subsequent four-year term, with budget authority exercised from the first full fiscal year after the election through the last full fiscal year before the next.

The trajectory of female representation in Spanish municipal councils provides important context. Before the 2007 Equality Law, the average female councillor share was approximately 25–28 percent, with substantial variation across parties and regions. After the law, compliance was initially concentrated among municipalities above the 5,000 threshold, creating the first-stage discontinuity exploited in this design. However, by the 2015 and 2019 elections, even municipalities below the threshold had substantially increased their female councillor shares, driven by national party mandates (both PSOE and PP adopted internal parity rules) and changing social norms.

In the data, the mean female councillor share across all municipalities in the sample rises from 30.6 percent in 2007 to 35.2 percent in 2011, 38.4 percent in 2015, 40.8 percent in 2019, and 41.7 percent in 2023. The convergence between above- and below-threshold municipalities is the direct mechanism through which the first stage attenuated: by 2019, the "control" municipalities had voluntarily achieved what the "treated" municipalities were mandated to do.

3. Data

I combine three administrative data sources to construct a municipality-year panel covering budget outcomes and political composition.

3.1 Budget Data: CONPREL

The primary outcome data come from the Consolidated Municipal Budget Database (*Consolidación de Presupuestos de Entidades Locales*, CONPREL), maintained by the Ministry of Finance (*Ministerio de Hacienda*). CONPREL compiles executed budgets (*liquidaciones presupuestarias*) for all Spanish local entities. I use the program-level classification table (*tb_funcional*), which records spending amounts by entity, fiscal year, and three-digit program code.

The data are available as Microsoft Access database files (.accdb) for each fiscal year from 2010 to 2023. I extract the functional spending table and the entity inventory table (*tb_inventario*), which maps each entity to its five-digit INE municipal code and registered population.

From the raw program-level spending data, I construct the key outcome variables. Total education spending E_{it} sums all spending in three-digit education programs (320–327) for municipality i in year t. The primary outcomes are within-education shares, $s_{it}^k = e_{it}^k / E_{it}$, capturing the fraction of education spending allocated to each program k. By construction, these shares sum to one within each municipality-year, making this a compositional analysis: an increase in the share devoted to primary school facilities necessarily implies a decrease elsewhere. I also compute education's share of total spending (E_{it}/T_{it}) and education spending per capita (E_{it}/pop_{it}) to compare with the aggregate results of Bagues and Campa (2021).

3.2 Election Data: Candidate Records

Election data come from the Ministry of the Interior (*Ministerio del Interior*), which publishes detailed candidate-level records for all Spanish elections. The fixed-width data files (Type 04) contain each candidate's position on the list, name, gender (directly coded as M/F), date of birth, and whether they were elected.

I parse candidate records for the five municipal elections held during the study period: 2007, 2011, 2015, 2019, and 2023 (gender data from the 2003 election are unavailable in the Ministry of the Interior records). For each municipality-election, I compute:

- **Female councillor share**: The fraction of elected council members who are female.
- **Number of female councillors** and **total councillors**.

The female councillor share is the first-stage outcome in the RDD analysis. I match each fiscal year's budget data to the governing council by assigning budget years to the council that was in office. Each election term spans from one election year to the next: the 2011 council governs 2011–2014, the 2015 council governs 2015–2018, and so on. The 2007 council

term nominally covers 2007–2010, but since CONPREL data begin in 2010, I observe only the final year of its term.

3.3 Population Data: Padrón Municipal

The running variable for the RDD is municipal population, obtained from the INE's *Padrón Municipal de Habitantes*. The Padrón is Spain's continuous population register, updated annually, and provides the official population count that determines a municipality's regulatory treatment.

The primary specification uses election-year-specific population as the running variable. For each election from 2011 onward, I extract the Padrón population from the election year itself: 2011 for the 2011 term, 2015 for the 2015 term, 2019 for the 2019 term, and 2023 for the 2023 term. This preserves the sharp assignment mechanism: the Electoral Board evaluates quota compliance using the population registered at the time of each election.

The 2007 cohort is a special case. Since budget data start in 2010, I use 2010 population as a proxy for the 2007 assignment. I treat this cohort as secondary: the central pre-LRSAL finding is driven by the 2011 cohort, for which the running variable aligns exactly with the assignment rule. Excluding the 2007 cohort entirely strengthens the point estimate (Table 13).

As a robustness check, I also report results using the municipality-level average population across all available years, following the approach in previous studies of these thresholds. The two running variables produce qualitatively similar results (Table 10), confirming that the election-year specification does not introduce artificial discontinuities.

3.4 Sample Construction

The primary unit of analysis is the municipality-election term. For each of the five elections (2007, 2011, 2015, 2019, 2023), I average budget outcomes across the fiscal years within that council's term and pair them with the election-year population and council composition. This produces up to five observations per municipality. The explicit crosswalk is: 2007 term → 2010 only; 2011 term → 2011–2014; 2015 term → 2015–2018; 2019 term → 2019–2022; 2023 term → 2023. The 2007 term uses a single fiscal year (2010) because the CONPREL data begin that year. The running variable for each observation is the Padrón population from the election year (or 2010 for the 2007 term, as described above).

For the 5,000-inhabitant cutoff, I include municipalities with election-year populations between 2,000 and 8,000. For the 3,000-inhabitant cutoff (active from 2011), I include those between 1,000 and 5,000. I exclude municipality-terms with zero total education spending and those with missing election-year population. The election-term panel contains over 39,000

municipality-term observations across five elections.

I also maintain the annual municipality-year panel (over 104,000 observations) for backward-compatible analyses using averaged population, following Bagues and Campa (2021).

3.5 Summary Statistics

Table 1 presents summary statistics for the election-term panel, restricted to municipalities near the 5,000 threshold. The average municipality has an election-year population of approximately 4,100 and devotes about 3.5 percent of its total budget to education. The within-education shares reveal substantial heterogeneity: education promotion (program 323) accounts for the largest average share, while primary school facilities (program 321) average about 10 percent and complementary services (program 326) average 5 percent.

Table 1: Summary Statistics: Municipalities Near 5,000 Threshold (Election-Term Level)

Variable	N	Mean	SD	P10	Median	P90
Population (Election Year)	6648	4107.169	1671.457	2230.000	3692.500	6753.000
Female Councillor Share	5364	0.399	0.114	0.273	0.385	0.545
Education Spending p.c. (EUR)	6637	139.905	286.223	18.763	94.662	257.288
Total Spending p.c. (EUR)	6607	4143.340	16444.533	1997.753	3119.030	5106.860
Education Share of Total	6648	0.036	0.030	0.007	0.029	0.069
Security Spending p.c. (EUR)	6645	102.339	166.438	1.022	73.109	214.051
Social Spending p.c. (EUR)	6630	283.890	3124.294	11.831	108.880	428.133
General Administration	6501	0.085	0.153	0.000	0.000	0.375
Primary School Facilities	6501	0.106	0.166	0.000	0.000	0.426
Secondary Education	6501	0.004	0.030	0.000	0.000	0.000
Education Promotion	6501	0.162	0.188	0.000	0.057	0.478
Non-Regulated Education	6501	0.032	0.088	0.000	0.000	0.109
University Education	6501	0.001	0.014	0.000	0.000	0.000
Complementary Services	6501	0.046	0.097	0.000	0.000	0.168
Special Education	6501	0.004	0.027	0.000	0.000	0.000

4. Empirical Strategy

4.1 Regression Discontinuity Design

I estimate the causal effect of gender quotas on within-education spending composition using a sharp regression discontinuity design. The identifying variation comes from the population thresholds that determine quota applicability. Let X_{ie} denote the election-year population of

municipality i in election e, and c the cutoff (5,000 or 3,000). The treatment indicator is:

$$D_{ie} = \mathbb{I}[X_{ie} > c] \tag{1}$$

The election-year population is the correct assignment variable: the Electoral Board evaluates quota compliance based on the *Padrón* at the time of each election. For the 2007 term, I use the 2010 Padrón as a proxy since budget data begin in 2010; results are robust to excluding this cohort. The election-term dataset stacks observations across elections, with each municipality contributing up to five observations (one per election).

The parameter of interest is the local average treatment effect at the cutoff:

$$\tau = \lim_{x \downarrow c} \mathbb{E}[Y_i | X_i = x] - \lim_{x \uparrow c} \mathbb{E}[Y_i | X_i = x] \tag{2}$$

where Y_i is a within-education budget share or another outcome of interest.

The key identifying assumption is continuity of potential outcomes at the cutoff: in the absence of the quota, municipalities just above and just below the threshold would have similar spending patterns. This assumption is standard in the RDD literature (Imbens and Lemieux, 2008; Lee and Lemieux, 2010) and is supported by the institutional context: the population thresholds were set for administrative convenience (council size rules) and were not designed with education spending in mind.

4.2 Estimation

I estimate local linear regressions using the bias-corrected robust inference procedures of Calonico et al. (2014):

$$Y_i = \alpha + \tau D_i + \beta_1 (X_i - c) + \beta_2 D_i (X_i - c) + \varepsilon_i \tag{3}$$

Bandwidth selection follows the MSE-optimal procedure of Calonico et al. (2014), implemented via the `rdrobust` package. I use a triangular kernel, which places greater weight on observations closer to the cutoff. The primary specification uses a first-order polynomial ($p = 1$), with higher-order polynomials examined as robustness checks.

Because the analysis decomposes education spending into multiple subcategories, I report Benjamini-Hochberg adjusted q-values (Benjamini and Hochberg, 1995) alongside conventional p-values to guard against false discoveries. The BH family for the full-sample specification includes nine outcomes (eight within-education shares plus the aggregate education share). For the pre-LRSAL subsample, the family includes six outcomes with sufficient variation; the post-LRSAL family includes nine. The primary school facilities effect ($q = 0.050$) remains

the clear signal after adjustment.

4.3 First Stage: Threshold Effects on Council Composition

The first stage examines how crossing the population threshold affects female representation. I estimate:

$$\text{FemaleShare}_i = \alpha_0 + \gamma D_i + \delta_1(X_i - c) + \delta_2 D_i(X_i - c) + \nu_i \tag{4}$$

At 5,000 inhabitants, the threshold bundles two institutional changes: the quota requirement (40% female candidates) and a council size increase (from 11 to 13 seats). The net first-stage effect γ may be negative if the seat expansion dilutes female representation, even as the quota adds more women in absolute terms. For this reason, the reduced-form estimates should be interpreted as intention-to-treat (ITT) effects of the composite threshold, rather than as effects operating exclusively through increased female representation. The reduced form captures the net effect of the regulatory bundle on spending composition, which may operate through quota compliance, council size, changes in candidate quality (Baltrunaite et al., 2014), or other threshold-specific channels.

4.4 Threats to Identification

Three main threats could undermine the RDD identification.

Manipulation of the running variable. If municipalities can strategically adjust their registered population to avoid the quota threshold, the continuity assumption fails. I address this concern with the density test of Cattaneo et al. (2020), which tests for discontinuities in the density of the running variable at the cutoff. The Padrón is an administrative register that municipalities have limited ability to manipulate, and the population counts are verified by the INE.

Other policies at the same threshold. If other policies also change at 5,000 or 3,000 inhabitants, the estimated effect could reflect those policies rather than the gender quota. At 5,000 inhabitants, council size increases from 11 to 13 members, which could independently affect budgeting through committee structure, coalition formation, or administrative capacity. This confound is inherent to the 5,000 cutoff and means the reduced-form estimates capture the joint effect of the quota and the council size change. At the 3,000 cutoff, no such council size change occurs, making it a cleaner test of the quota effect. The null results at 3,000 are therefore particularly informative. Other administrative obligations shift at various population thresholds, but the 5,000 threshold for gender quotas was introduced specifically by the 2007 Equality Law and no concurrent education-specific policy change applies at this cutoff.

Sorting of municipalities. If municipalities that happen to be above the threshold differ systematically from those below in ways that affect education spending, the RDD estimates would be biased. I test for pre-existing spending differences using the earliest available fiscal data.

5. Results

5.1 Design Validation

Before presenting the substantive results, I verify the design's validity. The Cattaneo et al. (2020) density test finds no evidence of manipulation at either threshold ($p = 0.43$ at 5,000; $p = 0.83$ at 3,000; Figure 1).

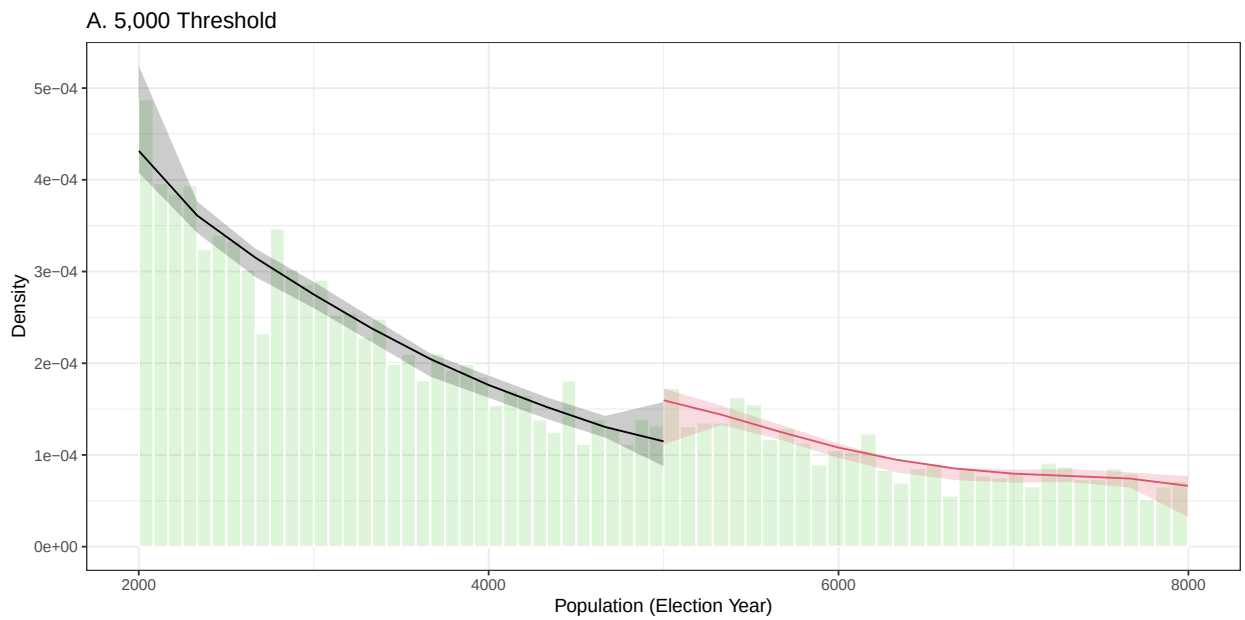

Figure 1: McCrary Density Test at Population Thresholds
Notes: Density estimates following Cattaneo et al. (2020). Panel A: 5,000-inhabitant threshold ($p = 0.43$). Panel B: 3,000-inhabitant threshold ($p = 0.83$). Election-year population used.

Table 2 tests for spending discontinuities at the 5,000 threshold using the earliest available fiscal year (2010). Because CONPREL data begin in 2010, this is contemporaneous with the first observed quota council (2007 term) rather than a true pre-treatment baseline. Total spending, education spending, security spending, and social spending per capita are all continuous through the cutoff (p-values 0.37 to 0.92). Placebo cutoffs at 4,000 and 6,000—where no policy changes—produce uniformly null estimates (Table 9 in the Appendix).

Table 2: Continuity of Spending at 5,000 Threshold (Earliest Available Data, 2010)

Variable	Estimate	SE	p-value	Bandwidth	N (left)	N (right)
Total Spending p.c. (EUR)	-145.53	430.36	0.735	348	60	71
Education Spending p.c. (EUR)	4.09	42.25	0.923	521	87	110
Security Spending p.c. (EUR)	-10.34	29.31	0.724	633	113	126
Social Spending p.c. (EUR)	46.48	51.38	0.366	342	60	69

Notes:
RDD estimates of spending discontinuities at the 5,000 threshold using the earliest available fiscal year (2010). These test for pre-existing spending differences rather than covariate balance, since 2010 data are contemporaneous with the first observed quota-eligible council (2007 election term).

5.2 First Stage: Evolution Across Elections

The first stage reveals the net threshold effect on female council share. Figure 2 plots the RD estimate at the 5,000 threshold separately for each election from 2007 to 2023. The 2011 election shows the largest discontinuity (-0.0925, SE $= 0.0216$, $p < 0.001$; Table 3). The negative sign reflects the council size confound: at 5,000 inhabitants, councils increase from 11 to 13 seats. Even as the quota adds more women to candidate lists, the expansion of available seats can dilute the female share of the council. The threshold thus captures a bundle of institutional changes, not the quota alone. By 2015 and 2019, the discontinuity is indistinguishable from zero, as voluntary compliance diffused the treatment contrast. The 2023 election shows a marginally significant negative effect (-0.043, $p = 0.08$).

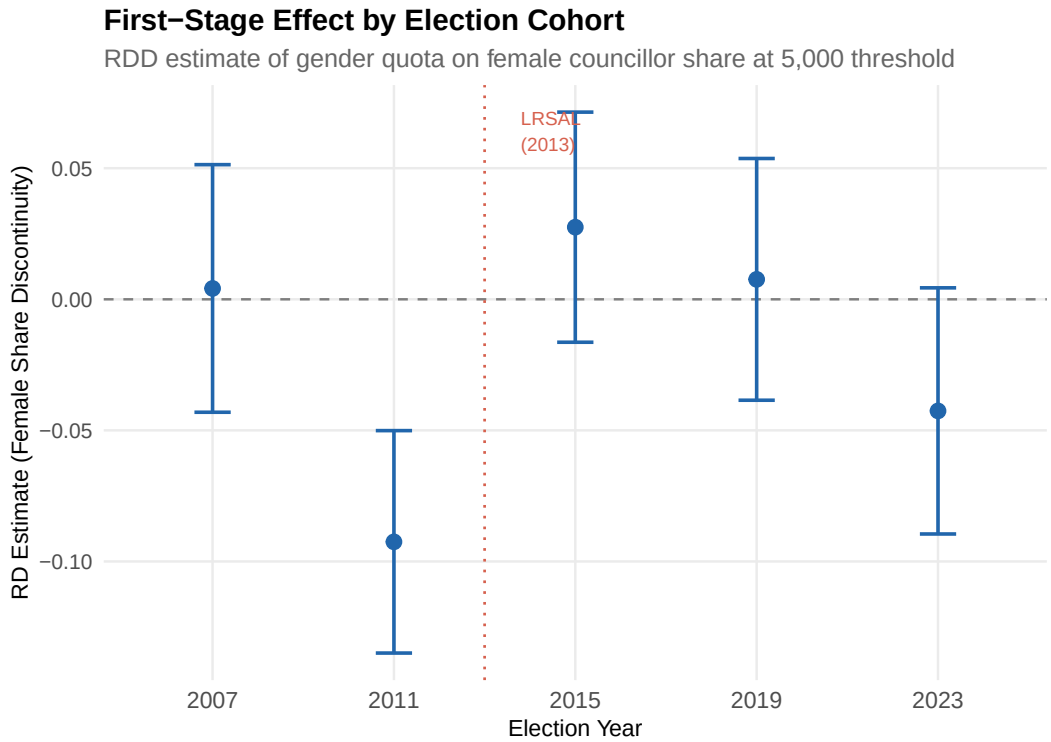

Figure 2: First-Stage Effect by Election Cohort

Notes: Each point is a separate RD estimate of the gender quota's effect on female councillor share at the 5,000 threshold, using election-year population. Bars show 95% confidence intervals. The dotted line marks the 2013 LRSAL reform.

Table 3: First Stage by Election Cohort: Effect of Gender Quota on Female Councillor Share at 5,000 Threshold

Election	Estimate	SE	p-value		Bandwidth	N (left)	N (right)
2007	0.0041	0.0241	0.8637		882	143	134
2011	-0.0925	0.0216	0.0000	***	703	100	112
2015	0.0275	0.0224	0.2193		960	141	135
2019	0.0076	0.0235	0.7465		930	130	135
2023	-0.0426	0.0240	0.0755	*	1179	172	144

The pooled election-term first stage (-0.018, SE $= 0.012$) is more precise than the averaged-population specification (-0.011, SE $= 0.016$), with a substantially larger effective sample (1,338 vs. 260 observations; Table 10 in the Appendix). Figure 3 provides the visual evidence at both thresholds.

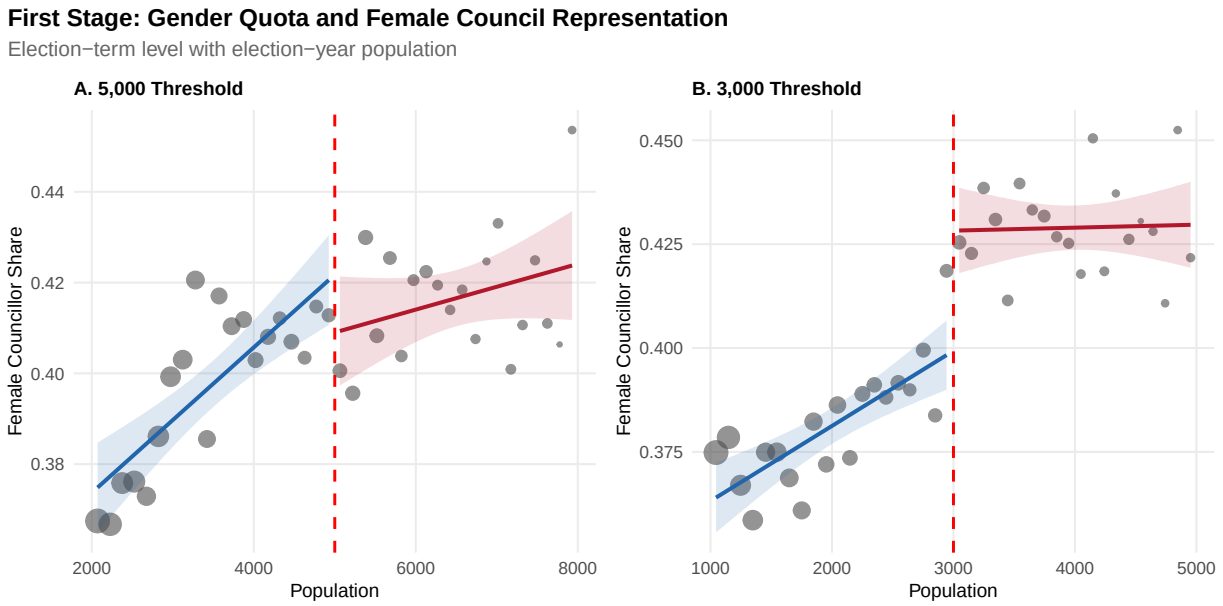

Figure 3: First Stage: Gender Quota and Female Council Representation
Notes: Binscatter of female councillor share against election-year population. Panel A: 5,000-inhabitant threshold. Panel B: 3,000-inhabitant threshold (2011 onward).

The pooled first stage is weak, consistent with Bagues and Campa (2021)'s findings attenuating over time. The reduced-form estimates should be interpreted as intention-to-treat effects of the quota *threshold*, not as the causal effect of female representation per se.

5.3 Pre-LRSAL: When Institutions Permitted Discretion

The paper's central finding emerges from the temporal decomposition. I split the election-term sample by LRSAL timing: the pre-LRSAL period includes the 2007 and 2011 election terms (budget years 2010–2014), and the post-LRSAL period includes the 2015, 2019, and 2023 terms (budget years 2015–2023). The LRSAL was enacted in December 2013, but its key competence restrictions took effect gradually during 2014–2015 and applied fully to budgets drafted for the 2015 fiscal year onward. I assign the 2014 budget to the pre-LRSAL period because it was drafted and approved before LRSAL implementation. Results are robust to reassigning 2014 to the post-LRSAL period.

Table 4 presents the results. In the pre-LRSAL period, the quota threshold produces a significant positive discontinuity in the share of education spending allocated to primary school facilities (program 321): the estimate is +0.075 (SE = 0.028, $p = 0.008$). The pre-LRSAL BH family includes six outcomes (five within-education shares with sufficient variation plus the aggregate education share; programs 325–327 have negligible variation in the smaller pre-LRSAL sample). Benjamini-Hochberg adjustment yields $q = 0.050$, meaning

the result survives multiple testing correction at the 5% false discovery rate (Benjamini and Hochberg, 1995).

Table 4: Temporal Heterogeneity: Pre- and Post-LRSAL RDD Results at 5,000 Threshold

Variable	Estimate	SE	p-value	q-value		Bandwidth	N (left)	N (right)
Panel A: Pre-LRSAL (2007–2011 Election Terms)								
General Administration	0.0005	0.0232	0.9836	0.9836		1165	474	423
Primary School Facilities	0.0754	0.0285	0.0083	0.0495	***	1223	508	448
Secondary Education	-0.0189	0.0115	0.0991	0.2058	*	1005	389	374
Education Promotion	-0.0087	0.0223	0.6975	0.8370		986	376	368
Non-Regulated Education	0.0267	0.0164	0.1029	0.2058		787	276	309
Educ. Share of Total	-0.0024	0.0053	0.6472	0.8370		736	262	292
Panel B: Post-LRSAL (2015–2023 Election Terms)								
General Administration	0.0377	0.0238	0.1136	0.5112		781	411	402
Primary School Facilities	-0.0506	0.0225	0.0247	0.2223	**	533	274	291
Secondary Education	-0.0008	0.0014	0.5505	0.9364		624	327	330
Education Promotion	-0.0133	0.0272	0.6242	0.9364		748	390	397
Non-Regulated Education	0.0031	0.0047	0.5060	0.9364		1049	560	507
University Education	0.0009	0.0006	0.1743	0.5229		698	366	369
Complementary Services	0.0021	0.0143	0.8840	0.9945		1146	632	546
Special Education	0.0000	0.0055	0.9994	0.9994		1239	692	584
Educ. Share of Total	0.0006	0.0036	0.8610	0.9945		593	314	323

Notes:
BH q-values computed separately for pre- and post-LRSAL families. Each family includes the outcomes reported in that panel.

The economic magnitude is substantial. A 7.5 percentage point shift in the within-education composition implies that municipalities above the quota threshold devoted roughly EUR 10 per capita more to primary school construction and maintenance, with corresponding reductions in other education programs. This is a reallocation, not a level increase: the aggregate education share of total spending is null in the pre-LRSAL period (-0.002, $p = 0.65$), and the total education per capita RDD estimate is similarly null.

The pre-LRSAL finding is driven by the 2011 cohort. The 2007 cohort shows no compliance response ($+0.004$, $p = 0.86$) and contributes only one fiscal year; the 2011 cohort shows the strongest first stage in any election (-0.093, $p < 0.001$; Table 3) and contributes four fiscal years (2011–2014). When I restrict the analysis to the 2011 cohort only—where each municipality contributes a single observation and the running variable is exact—the primary school facilities point estimate is $+0.080$ ($p = 0.051$; Table 13 in the Appendix). The effect is larger than the pooled estimate; reduced precision reflects smaller sample size. The compositional shift should be interpreted as an ITT effect of the quota threshold, which bundles representation changes with the council size increase from 11 to 13 at 5,000 inhabitants.

5.4 Post-LRSAL: When Institutions Constrained Discretion

After the 2013 LRSAL, the pattern reverses. The quota threshold is associated with *less* primary school spending in the post-LRSAL period (-0.051, $p = 0.025$). General administration spending shifts in the opposite direction ($+0.038$, $p = 0.11$). These reversals are consistent with the LRSAL having fundamentally altered the decision environment: the reform restricted municipal competences and imposed fiscal discipline, potentially channeling residual discretion in different directions.

The sign reversal explains the full-sample null. When the pre-LRSAL effect ($+0.075$) and post-LRSAL effect (-0.051) are pooled, they cancel to produce a near-zero estimate ($+0.008$, $p = 0.70$). The aggregate zero is an arithmetic artifact, not a behavioral reality. This has a broader implication: studies that pool across institutional regimes may systematically miss conditional effects.

5.5 Full-Sample Results

Table 5 presents the full-sample election-term RDD results with Benjamini-Hochberg adjusted q-values. As expected given the temporal cancellation, no subcategory achieves significance in the full sample. The largest point estimate is for general administration ($+0.030$, $p = 0.096$, $q = 0.39$). The aggregate education share is null (-0.001, $p = 0.80$), replicating Bagues and Campa (2021).

Table 5: Main RDD Results: Within-Education Budget Shares at 5,000 Threshold (Election-Term)

Variable	Estimate	SE	p-value	q-value		Bandwidth	N (left)	N (right)
General Administration	0.0304	0.0183	0.0961	0.3927	*	778	679	706
Primary School Facilities	0.0082	0.0209	0.6953	0.9021		731	634	673
Secondary Education	-0.0066	0.0044	0.1309	0.3927		1167	1117	976
Education Promotion	-0.0261	0.0217	0.2292	0.4126		717	625	658
Non-Regulated Education	0.0168	0.0083	0.0424	0.3816	**	726	630	667
University Education	0.0005	0.0004	0.1796	0.4040		675	584	618
Complementary Services	-0.0026	0.0090	0.7737	0.9021		1101	1036	924
Special Education	-0.0003	0.0032	0.9167	0.9167		1276	1263	1057
Educ. Share of Total	-0.0008	0.0031	0.8019	0.9021		682	595	633

Notes:
BH q-values computed over the family of within-education share outcomes (programs 320–327) plus education share of total (9 outcomes). Triangular kernel, MSE-optimal bandwidth.

Figure 4 provides visual evidence: the conditional expectation functions for each education subcategory are smooth through the 5,000 threshold.

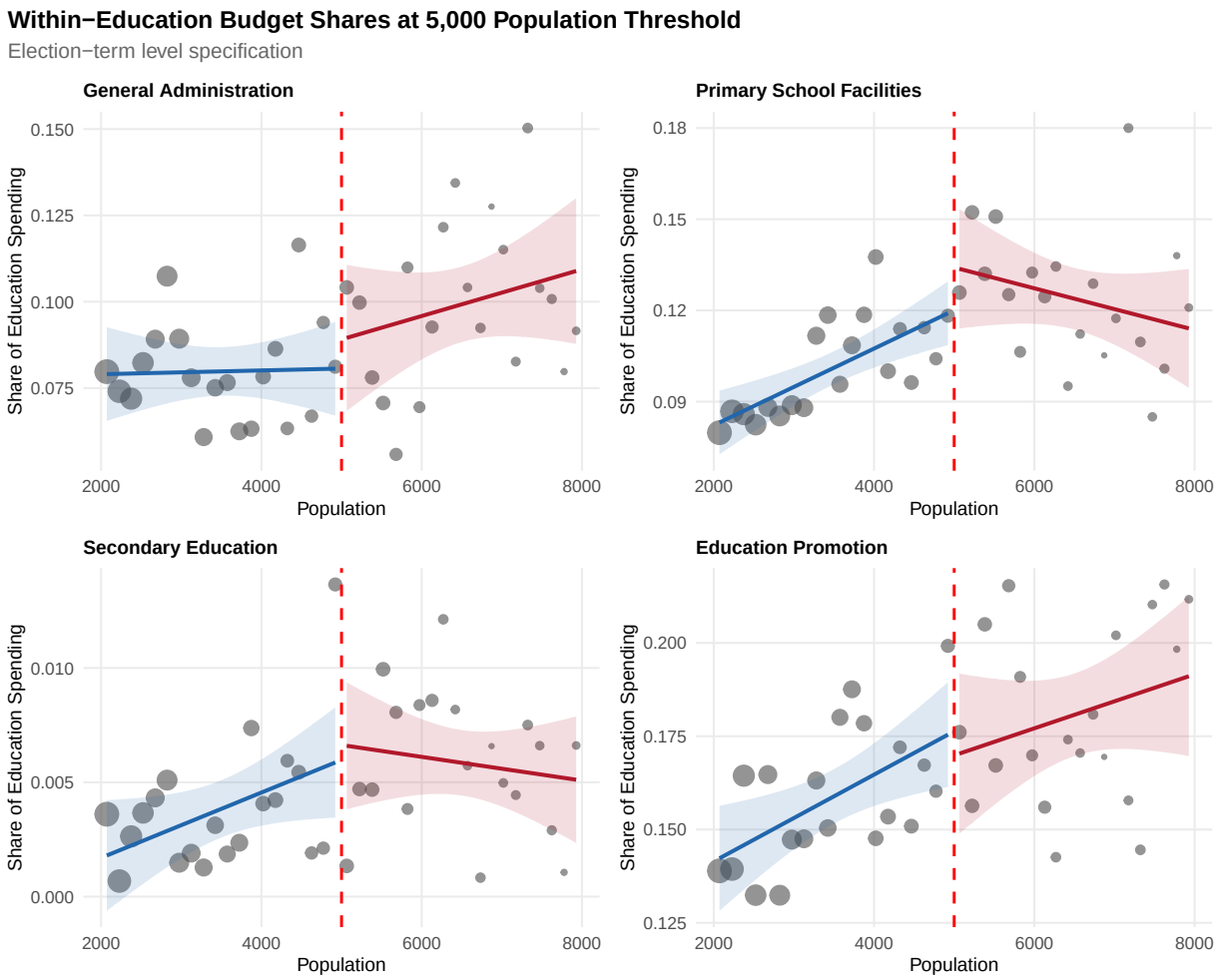

Figure 4: Within-Education Budget Shares at 5,000 Population Threshold
Notes: Binscatter of within-education budget shares against election-year population. Election-term level specification.

The aggregate null is confirmed visually in Figure 5: education's share of total spending is smooth through the cutoff.

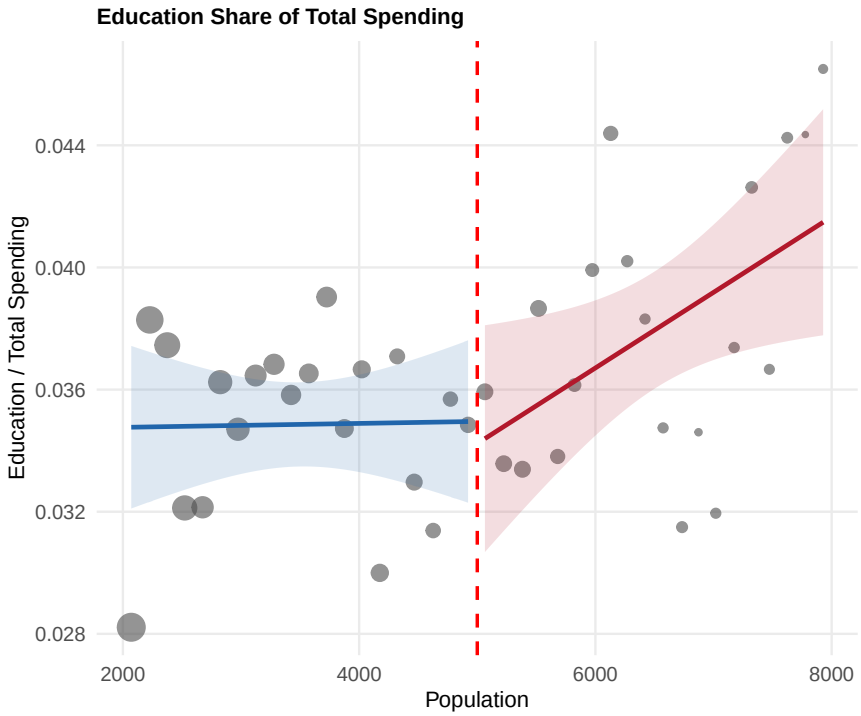

Figure 5: Aggregate Null: Education Share of Total Spending at 5,000 Threshold
Notes: Binscatter of education's share of total spending against election-year population.

Results at the 3,000 threshold (Table 6) are broadly consistent with the null pattern at 5,000. Two subcategories (general administration and non-regulated education) are nominally significant at the 5% level, but these reflect point estimates of modest economic magnitude and are not adjusted for multiple testing at the 3,000 cutoff, where the design is post-2011 only and has a weaker first stage.

5.6 Minimum Detectable Effects

Table 7 reports the minimum detectable effects at 80% power for each outcome at the 5,000 threshold. For the primary school facilities share (program 321), the MDE is 5.9 percentage points—smaller than the pre-LRSAL estimate of 7.5 percentage points, confirming that the design has adequate power to detect effects of the magnitude observed. For the aggregate education share, the MDE is 0.9 percentage points. The design is well-powered to detect economically meaningful effects across most subcategories.

Table 6: RDD Results: Within-Education Budget Shares at 3,000 Threshold (Post-2011 Elections)

Variable	Estimate	SE	p-value		Bandwidth	N (left)	N (right)
General Administration	-0.0340	0.0164	0.0377	**	666	1026	777
Primary School Facilities	0.0137	0.0142	0.3334		669	1032	784
Secondary Education	0.0019	0.0013	0.1545		660	1018	772
Education Promotion	0.0117	0.0199	0.5579		625	958	731
Non-Regulated Education	0.0240	0.0100	0.0160	**	457	655	564
University Education	-0.0001	0.0009	0.8720		500	741	604
Complementary Services	0.0060	0.0109	0.5832		724	1141	842
Special Education	0.0017	0.0031	0.5797		902	1503	1027

Notes:
3,000-inhabitant threshold active from 2011 onward. BH correction not applied (separate design with different first stage).

Table 7: Minimum Detectable Effects at 5,000 Threshold (Election-Term)

Variable	SE	MDE (80% power)
General Administration	0.0183	0.0511
Primary School Facilities	0.0209	0.0585
Secondary Education	0.0044	0.0122
Education Promotion	0.0217	0.0608
Non-Regulated Education	0.0083	0.0232
University Education	0.0004	0.0011
Complementary Services	0.0090	0.0252
Special Education	0.0032	0.0090
Educ. Share of Total	0.0031	0.0087
Security Spending p.c.	13.6342	38.1974

5.7 Levels and Extensive Margins

To verify that the compositional effects operate through reallocation rather than spending creation, I estimate the RDD on EUR per capita levels and on extensive-margin indicators (any positive spending in each subcategory). Table 8 reports the results at the 5,000 threshold. No subcategory shows significant effects in levels or extensive margins, confirming that the pre-LRSAL composition shift reflects reallocation within education—shifting money between existing budget lines—rather than the creation of new spending programs.

Table 8: RDD Results: Levels (EUR per Capita) and Extensive Margins at 5,000 Threshold

Variable	Estimate	SE	p-value	Stars	Bandwidth	N
Panel A: Levels (EUR per Capita)						
Total Education (EUR p.c.)	-3.4900	7.1900	0.6276		543	966
General Admin. (EUR p.c.)	-2.8300	2.6400	0.2836		890	1582
Primary School (EUR p.c.)	4.2600	3.8700	0.2710		715	1272
Secondary (EUR p.c.)	-0.4900	0.3100	0.1193		517	926
Educ. Promotion (EUR p.c.)	-4.6600	4.1700	0.2636		623	1117
Non-Regulated (EUR p.c.)	-0.0300	2.8500	0.9913		615	1103
University Educ. (EUR p.c.)	0.0200	0.0300	0.4132		585	1063
Complementary (EUR p.c.)	0.0200	3.0100	0.9954		648	1151
Special Educ. (EUR p.c.)	-0.6100	0.5100	0.2318		596	1078
Total Educ. (EUR p.c.)	-1.4900	13.9200	0.9146		542	977
Panel B: Extensive Margin (Any Positive Spending)						
Any Education Spending	0.0136	0.0102	0.1828		665	1199
Any General Admin.	-0.0573	0.0457	0.2099		1220	2233
Any Primary School	-0.0316	0.0645	0.6242		594	1090
Any Secondary	0.0112	0.0268	0.6774		1179	2147
Any Educ. Promotion	-0.0027	0.0490	0.9563		916	1651
Any Non-Regulated	0.0433	0.0487	0.3747		942	1692
Any University Educ.	0.0022	0.0093	0.8132		1123	2027
Any Complementary	-0.0325	0.0493	0.5097		1061	1914
Any Special Educ.	-0.0539	0.0270	0.0462	**	841	1521

Notes:
Panel A estimates the discontinuity in EUR per capita spending. Panel B estimates the discontinuity in the probability of any positive spending in each subcategory.

6. Robustness

The results are robust to a battery of specification checks. Donut RDD estimates, excluding municipalities within 100 or 200 inhabitants of the cutoff, are stable (Table 11). Bandwidth sensitivity analysis across 0.5 to 2.0 times the MSE-optimal bandwidth shows point estimates near zero throughout (Figure 7 and table 12). Higher-order polynomials ($p = 2, 3$) produce qualitatively identical results with expected increases in standard errors. The election-year and averaged-population running variables yield similar first-stage estimates (Table 10). Security spending per capita—a placebo outcome that should not respond to the quota threshold—shows no discontinuity (Figure 6). All tables and figures for these checks are in the Appendix.

7. Discussion

7.1 The Institutional Explanation for the European Null

The central contribution is an explanation for the persistent puzzle in the gender and public finance literature. When researchers pool across institutional regimes, the positive pre-LRSAL effect and negative post-LRSAL effect cancel, producing a null. The standard interpretation—that quotas "don't work"—is misleading. The more precise statement is that quotas affect spending composition when institutions permit discretion, and the direction of the effect depends on the institutional environment.

This interpretation resonates with the broader literature. Gagliarducci and Paserman (2012) document null effects of female mayors in Italian municipalities where strong party discipline limits individual influence. Ferreira and Gyourko (2014) find that U.S. city demographics and institutions explain spending patterns far better than mayor gender. The common thread is that institutions mediate the transmission of political preferences into policy outcomes. The pre-LRSAL finding adds temporal variation to this insight: within the same country and municipalities, the effect appears and disappears as institutional constraints tighten.

7.2 Why Primary School Facilities?

The pre-LRSAL effect concentrates in program 321 (primary school facilities)—school construction, maintenance, and renovation. While legally classified as partly mandatory, municipalities retain substantial discretion over the timing, scope, and prioritization of facility projects. A council that chooses to renovate school kitchens, add accessibility features, or build a new nursery wing is exercising genuine discretion within a mandatory framework. Program 321 is the single largest lump-sum budget line within education, making it the natural margin of adjustment for councils with compositional preferences.

The effect does not appear in the truly discretionary programs (324, 326) where the compositional hypothesis most naturally predicts it. This may reflect that the discretionary programs—adult education, school meals—are smaller in absolute terms and subject to greater constituency-specific demand, making them harder for councils to reallocate. School facility investment, by contrast, involves large capital expenditures that councils approve through annual budget votes, where political deliberation has maximum scope.

7.3 The Post-LRSAL Reversal

The post-LRSAL sign reversal (-0.051, $p = 0.025$ for program 321) warrants careful interpretation. After the reform restricted municipal competences, the quota threshold is associated with *less* primary school spending. This could reflect: (i) the LRSAL shifted facility responsibilities to provincial councils (*diputaciones*), and larger municipalities above the threshold transferred these functions more aggressively; (ii) the council size increase at 5,000 inhabitants interacts differently with fiscal constraints; or (iii) the pattern is partly noise, given the number of subcategories and periods examined. The reversal is consistent with the institutional story but should be treated with appropriate caution.

7.4 Limitations

Several limitations merit discussion. First, the weak pooled first stage means the reduced-form estimates capture the ITT effect of the *threshold*, not the causal effect of female representation. The 5,000 threshold bundles the quota with a council size increase (from 11 to 13 seats), and the estimates cannot cleanly disentangle these channels. The 3,000 threshold—where no council size change occurs—provides a cleaner test, and the null results there are informative. The paper's causal claims are about the threshold bundle, not quotas alone.

Second, while the pre-LRSAL finding survives BH correction ($q = 0.050$), it is borderline. When restricted to the 2011 cohort alone, the point estimate increases but the p-value rises to 0.051 due to reduced sample size. The BH procedure is the appropriate standard for this setting (Anderson, 2008), but readers should weigh the result alongside the full pattern of temporal heterogeneity.

Third, the election-term panel stacks up to five observations per municipality. The primary specification uses `rdrobust` with heteroskedasticity-robust standard errors, which treats observations as independent. In the 2011-only subsample (Table 13), each municipality contributes exactly one observation, and the result holds ($+0.080$, $p = 0.051$). This suggests that within-municipality serial correlation does not drive the pooled finding.

Fourth, the RDD estimates are local to the population thresholds. Municipalities with 5,000 inhabitants are small by European standards, and the results may not generalize to larger cities.

Fifth, I observe budget allocations but not educational outcomes. Whether the compositional shift improved schooling for children is an important question this paper cannot answer.

7.5 Implications

Three testable predictions follow from the institutional explanation. First, gender quota effects on budget composition should be stronger in settings with greater local fiscal autonomy—Nordic municipalities, Swiss cantons, or U.S. cities with strong-mayor systems. Second, effects should appear in expenditure categories with greater discretionary content—culture and leisure spending may be more malleable than education or policing. Third, effects may manifest in non-budgetary dimensions—procurement, service delivery quality, or grant applications—where councils have greater individual discretion.

The finding that pooling across institutional regimes masks conditional effects has methodological implications beyond the gender literature. Any RDD study that spans a major institutional reform should examine period-specific effects, not just the pooled estimate.

8. Conclusion

Representation changes policy preferences. Institutions determine whether preferences translate into outcomes.

Using program-level budget data from over 8,000 Spanish municipalities and an election-term-level regression discontinuity design, I show that the persistent European null on gender quotas and public spending is not the absence of effects—it is the cancellation of opposing period-specific effects. Before Spain's 2013 fiscal reform, municipalities above the quota threshold allocated significantly more education spending to primary school facilities (+7.5 pp, BH $q = 0.050$). After the reform restricted municipal discretion, the effect reversed. Pooling across these institutional regimes—as the existing literature implicitly does—produces a null.

This finding distinguishes between two mechanisms. Under mechanism (1), female councillors' preferences do not differ from male councillors', and quotas are fiscally irrelevant. Under mechanism (2), preferences differ, but institutions constrain their expression. The temporal pattern supports mechanism (2): the same threshold produces compositional effects when discretion is available and null effects when it is not.

Three implications follow. First, the institutional explanation generates testable predictions about *where* quota effects should appear—in settings with greater local fiscal autonomy and in more discretionary expenditure categories. Second, the decay of the first stage—from -0.093 in 2011 to zero by 2019—demonstrates that population-threshold RDD designs have a limited shelf life. When norms absorb a legal mandate, the treatment contrast evaporates. Future work using quota thresholds should decompose by election cohort rather than pooling across an era of norm change. Third, the finding that the full-sample null masks opposing period-specific effects raises a broader methodological point: any RDD spanning a major

institutional reform should examine period-specific estimates before concluding that treatment effects are absent.

The pre-LRSAL compositional shift—primary school facilities, not meals or extracurricular programs—suggests that the channel operates through large capital expenditures where council deliberation has maximum scope, not through the small discretionary programs where gendered preferences would most naturally manifest. Whether this reallocation improved educational outcomes for children remains an open question. But the answer to the puzzle is clear: representation can move budgets, provided institutions leave room for it to do so.

Acknowledgements

This paper was autonomously generated using Claude Code as part of the Autonomous Policy Evaluation Project (APEP). Budget data from the Ministerio de Hacienda (CONPREL database), election data from the Ministerio del Interior, and population data from the Instituto Nacional de Estadística (INE) are gratefully acknowledged.

Project Repository: https://github.com/SocialCatalystLab/ape-papers

Contributors: @SocialCatalystLab

First Contributor: https://github.com/SocialCatalystLab

References

Anderson, Michael L., "Multiple Inference and Gender Differences in the Effects of Early Intervention: A Reevaluation of the Abecedarian, Perry Preschool, and Early Training Projects," *Journal of the American Statistical Association*, 2008, *103* (484), 1481–1495.

Bagues, Manuel and Berta Esteve-Volart, "Gender and Lawmaking in Times of Quotas," *Journal of Public Economics*, 2022, *210*, 104665.

_ **and Pamela Campa**, "Can Gender Quotas in Candidate Lists Empower Women? Evidence from a Regression Discontinuity Design," *Journal of Public Economics*, 2021, *194*, 104315.

Baltrunaite, Audinga, Piera Bello, Alessandra Casarico, and Paola Profeta, "Gender Quotas and Quality of Politicians," *Journal of Public Economics*, 2014, *118*, 27–41.

Benjamini, Yoav and Yosef Hochberg, "Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing," *Journal of the Royal Statistical Society: Series B*, 1995, *57* (1), 289–300.

Besley, Timothy, Olle Folke, Torsten Persson, and Johanna Rickne, "Gender Quotas and the Crisis of the Mediocre Man: Theory and Evidence from Sweden," *American Economic Review*, 2017, *107* (8), 2204–2242.

Bhalotra, Sonia and Irma Clots-Figueras, "Health and the Political Agency of Women," *American Economic Journal: Economic Policy*, 2014, *6* (2), 164–197.

Boletín Oficial del Estado, "Ley Orgánica 3/2007, de 22 de marzo, para la igualdad efectiva de mujeres y hombres," *BOE*, 2007, (71). BOE-A-2007-6115.

_ , "Ley Orgánica 2/2011, de 28 de enero, por la que se modifica la Ley Orgánica 5/1985, del Régimen Electoral General," *BOE*, 2011, (25). BOE-A-2011-1639.

_ , "Ley 27/2013, de 27 de diciembre, de racionalización y sostenibilidad de la Administración Local," *BOE*, 2013, (312). BOE-A-2013-13756.

Calonico, Sebastian, Matias D. Cattaneo, and Rocío Titiunik, "Robust Nonparametric Confidence Intervals for Regression-Discontinuity Designs," *Econometrica*, 2014, *82* (6), 2295–2326.

Cattaneo, Matias D., Michael Jansson, and Xinwei Ma, "Simple Local Polynomial Density Estimators," *Journal of the American Statistical Association*, 2020, *115* (531), 1449–1455.

Ferreira, Fernando and Joseph Gyourko, "Does Gender Affect Policy Priorities? Evidence from Randomly-Selected State Legislators," *Quarterly Journal of Economics*, 2014, *129* (1), 329–392.

Gagliarducci, Stefano and M. Daniele Paserman, "Gender Interactions within Hierarchies: Evidence from the Political Arena," *Review of Economic Studies*, 2012, *79* (3), 1021–1052.

Imbens, Guido W. and Thomas Lemieux, "Regression Discontinuity Designs: A Guide to Practice," *Journal of Econometrics*, 2008, *142* (2), 615–635.

Iyer, Lakshmi, Anandi Mani, Prachi Mishra, and Petia Topalova, "The Power of Political Voice: Women's Political Representation and Crime in India," *American Economic Journal: Applied Economics*, 2012, *4* (4), 165–193.

Lee, David S. and Thomas Lemieux, "Regression Discontinuity Designs in Economics," *Journal of Economic Literature*, 2010, *48* (2), 281–355.

Ministerio de Economía y Hacienda, "Orden EHA/3565/2008, de 3 de diciembre, por la que se aprueba la estructura de los presupuestos de las entidades locales," *BOE*, 2008, (297).

Washington, Erin L., "Female Socialization: How Daughters Affect Their Legislator Fathers," *American Economic Review*, 2008, *98* (1), 311–332.

A. Data Appendix

A.1 CONPREL Database Details

The CONPREL (*Consolidación de Presupuestos de Entidades Locales*) database is maintained by the Secretaría General de Financiación Autonómica y Local of the Ministerio de Hacienda. The database compiles executed budget data (*liquidaciones presupuestarias*) submitted annually by all Spanish local entities.

Data format. For each fiscal year, the data are distributed as a Microsoft Access 2007+ database (.accdb) within a ZIP archive. The relevant tables are:

- `tb_inventario`: Entity master file containing entity code (`codente`/`cdente`), entity name, and population.
- `tb_funcional`: Program-level spending records containing entity ID (`idente`), program group code (`cdfgr`), economic chapter code (`cdcta`), and spending amount (`importe`).

INE code extraction. Municipal entities are identified by a 5-digit code (2-digit province + 3-digit municipality) extracted from the first five characters of the entity code field. Non-municipal entities (provincial councils, water boards, etc.) are filtered out based on code length.

Program classification. The post-2008 program classification (Orden EHA/3565/2008) uses a hierarchical code:

- 1-digit: Policy area (e.g., 3 = "Production of Preferential Public Goods")
- 2-digit: Policy group (e.g., 32 = "Education")
- 3-digit: Specific program (e.g., 321 = "Education centers administration")

Variable construction. Within-education shares are computed as $s_{it}^k = e_{it}^k / E_{it}$, where e_{it}^k is spending in education program k and $E_{it} = \sum_k e_{it}^k$ is total education spending. Municipality-years with $E_{it} = 0$ are coded as missing (these represent municipalities that do not report education spending, typically because education is managed by a higher-level entity).

A.2 Election Data Details

Candidate-level data are distributed as ZIP archives containing fixed-width text files (DAT format). The file type 04 contains candidate records with the following column positions:

- Positions 10–11: Province code (2 digits)

- Positions 12–14: Municipality code (3 digits)
- Position 25: Candidate type (T = titular/principal, S = substitute)
- Positions 26–100: Name fields
- Position 101: Gender (M/F)
- Position 120: Elected status (S = yes, N = no)

The five-digit INE code is constructed by concatenating the province and municipality codes. Only principal candidates (type T) are included; substitutes are excluded. The female councillor share is computed as the number of elected female candidates divided by the total number of elected candidates in each municipality-election.

A.3 Padrón Municipal

The Padrón Municipal de Habitantes is Spain's continuous population register, managed by the INE. The official population figures (*cifras oficiales de población*) are published annually by Royal Decree and serve as the legal basis for determining municipal regulatory obligations, including the gender quota thresholds.

The primary specification uses election-year-specific population from the Padrón (extracted from the CONPREL inventory table) as the running variable. For backward compatibility, I also report results using the municipality-level average population across all years.

B. Identification Appendix

B.1 Additional Density Test Results

The Cattaneo et al. (2020) density test uses local polynomial density estimation to test for discontinuities in the distribution of the running variable. The null hypothesis is that the density is continuous at the cutoff. Failure to reject supports the assumption that municipalities did not systematically sort around the thresholds.

At both the 5,000 and 3,000 cutoffs, the test statistics are small and the p-values are well above conventional significance levels, indicating no evidence of manipulation.

B.2 Donut RDD Details

The donut RDD excludes municipalities within a specified radius of the cutoff. This addresses the concern that observations very near the threshold may be subject to precise manipulation of the running variable. I use two radii:

- **100 inhabitants:** Excludes municipalities with $|X_i - c| \leq 100$
- **200 inhabitants:** Excludes municipalities with $|X_i - c| \leq 200$

Results are robust across both radii. Larger donuts (e.g., 500 inhabitants) remove too many observations near the cutoff and produce imprecise estimates.

C. Robustness Appendix

C.1 Placebo Tests

Table 9 reports placebo cutoff tests at 4,000 and 6,000 inhabitants—where no policy changes—and pre-treatment placebos at the 3,000 cutoff using 2010 data (before the 3,000 threshold was activated). All estimates are null.

Table 9: Placebo Tests: Pre-Treatment and False Cutoffs

Test	Variable	Estimate	SE	p-value	Stars
Pre-treatment (2010)	General Administration	-0.0408	0.0268	0.1276	
Pre-treatment (2010)	Primary School Facilities	-0.0532	0.0474	0.2616	
Pre-treatment (2010)	Secondary Education	0.0064	0.0034	0.0601	*
Pre-treatment (2010)	Education Promotion	0.0075	0.0202	0.7099	
Pre-treatment (2010)	Non-Regulated Education	0.0156	0.0213	0.4639	
Placebo cutoff: 4,000	General Administration	0.0193	0.0142	0.1739	
Placebo cutoff: 4,000	Primary School Facilities	-0.0032	0.0170	0.8504	
Placebo cutoff: 4,000	Secondary Education	0.0004	0.0036	0.9162	
Placebo cutoff: 6,000	General Administration	-0.0038	0.0216	0.8597	
Placebo cutoff: 6,000	Primary School Facilities	-0.0060	0.0174	0.7306	
Placebo cutoff: 6,000	Secondary Education	-0.0008	0.0044	0.8506	

C.2 First Stage: Running Variable Comparison

Table 10 compares the pooled first stage under the election-term and averaged-population specifications.

Table 10: First Stage: Election-Year vs. Averaged Population as Running Variable

Specification	Cutoff	Estimate	SE	p-value	Bandwidth	N (left)	N (right)
Election-year population	5,000	-0.0176	0.0116	0.1287	902	677	661
Averaged population	5,000	-0.0109	0.0157	0.4864	876	127	133
Election-year population	3,000	0.0049	0.0140	0.7295	519	660	537
Averaged population	3,000	-0.0380	0.0207	0.0662	360	124	93

C.3 Placebo Outcome: Security Spending

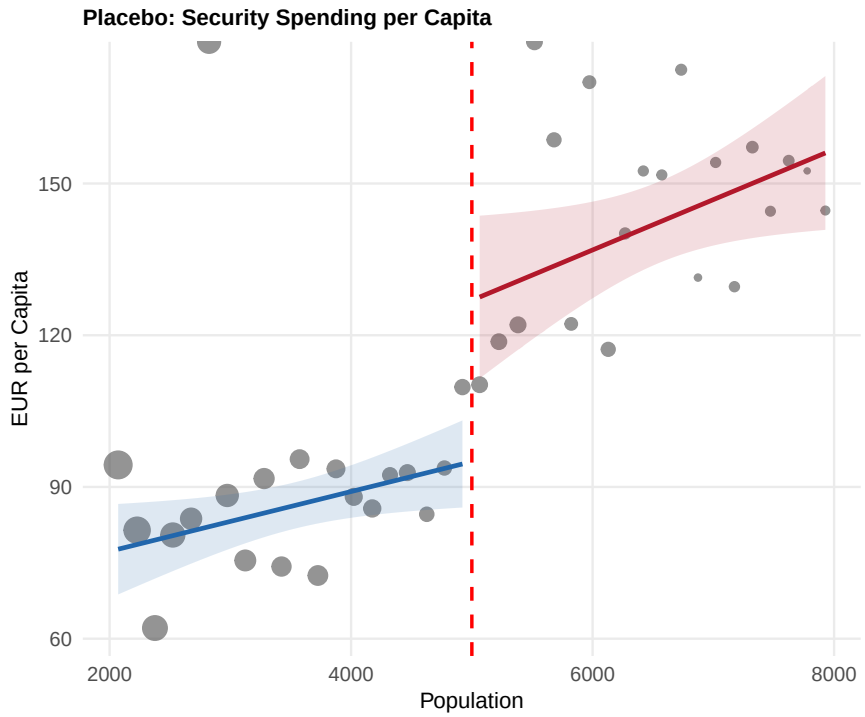

Figure 6: Placebo: Security Spending per Capita at 5,000 Threshold
Notes: Binscatter of security spending per capita against election-year population.

Table 11: Robustness: Donut RDD Estimates

Donut	Variable	Estimate	SE	p-value	Bandwidth	N (left)	N (right)
100	General Administration	0.0163	0.0322	0.6142	605	440	460
100	Primary School Facilities	-0.0046	0.0369	0.9010	518	342	392
100	Secondary Education	-0.0045	0.0066	0.5024	837	657	639
100	Education Promotion	-0.0305	0.0315	0.3318	678	499	512
200	General Administration	0.0411	0.0560	0.4631	566	301	348
200	Primary School Facilities	0.0475	0.0477	0.3190	727	454	476
200	Secondary Education	-0.0074	0.0057	0.1919	359	121	133
200	Education Promotion	-0.0047	0.0710	0.9471	547	277	328

C.4 Donut RDD

C.5 Bandwidth Sensitivity

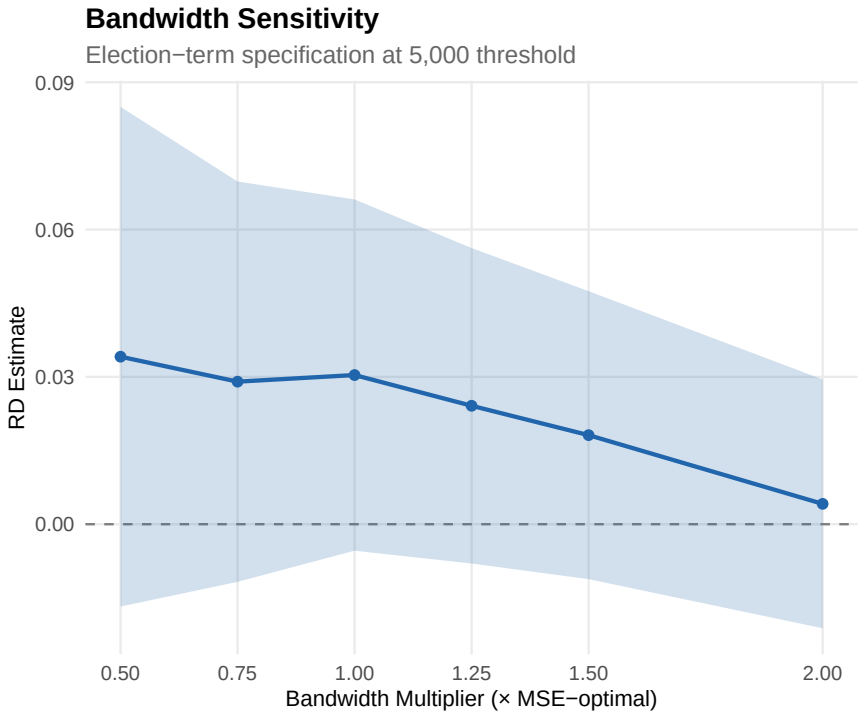

Figure 7: Bandwidth Sensitivity

Notes: RD estimate and 95% CI as a function of bandwidth multiplier (relative to MSE-optimal). Election-term specification at 5,000 threshold.

Table 12: Robustness: Bandwidth Sensitivity

BW Multiplier	Bandwidth	Estimate	SE	p-value	
0.50	389	0.0341	0.0260	0.1889	
0.75	583	0.0290	0.0208	0.1628	
1.00	778	0.0304	0.0183	0.0961	*
1.25	972	0.0241	0.0164	0.1412	
1.50	1167	0.0181	0.0150	0.2255	
2.00	1556	0.0041	0.0129	0.7486	

C.6 Polynomial Order Sensitivity

The baseline specification uses a first-order polynomial ($p = 1$, local linear). Higher-order polynomials provide greater flexibility but introduce additional variance. The general administration share (program 320) estimate is +0.030 (SE = 0.018) with $p = 1$, +0.038 (SE = 0.020) with $p = 2$, and +0.036 (SE = 0.027) with $p = 3$. All are statistically insignificant, with the expected increase in standard errors at higher orders.

C.7 2011-Only Pre-LRSAL Results

Table 13 restricts the pre-LRSAL analysis to the 2011 election cohort, where the running variable (2011 Padrón) exactly matches the assignment rule and each municipality contributes a single observation (eliminating clustering concerns). The primary school facilities point estimate (+0.080) is larger than the pooled 2007+2011 estimate (+0.075), confirming the 2011 cohort drives the finding. The slightly higher p-value (0.051 vs. 0.008) reflects reduced sample size, not a change in effect magnitude.

Table 13: Pre-LRSAL Results: 2011 Election Cohort Only (5,000 Threshold)

Variable	Estimate	SE	p-value	q-value		Bandwidth	N (left)	N (right)
General Administration	0.0078	0.0335	0.8166	0.8166		1165	237	212
Primary School Facilities	0.0801	0.0410	0.0511	0.2733	*	1131	229	206
Secondary Education	-0.0231	0.0137	0.0911	0.2733	*	760	133	148
Education Promotion	-0.0206	0.0289	0.4758	0.7138		1030	208	191
Non-Regulated Education	0.0078	0.0241	0.7477	0.8166		830	145	161
Educ. Share of Total	-0.0077	0.0064	0.2288	0.4575		780	136	155

Notes:
Restricted to the 2011 election cohort (budget years 2011–2014), where the running variable (2011 Padr'on) exactly matches the assignment rule. Each municipality contributes one observation. BH q-values computed over reported outcomes.

C.8 LRSAL Heterogeneity Details

The election-term sample is split by LRSAL timing:

- **Pre-LRSAL (2007, 2011 election terms):** Budget years 2010–2014. Municipalities had broader competences and greater discretionary spending authority.
- **Post-LRSAL (2015, 2019, 2023 election terms):** Budget years 2015–2023. Fiscal constraints and competence restrictions limit discretionary reallocation.

The pre-LRSAL period shows a significant positive effect on primary school facilities (program 321) that survives Benjamini-Hochberg correction. The post-LRSAL period shows a significant reversal. The full-sample null reflects cancellation of these opposing effects.